

L26 Compiler Design Report

Implementation: Flex lexer, Bison parser and semantic actions, P-Code generator, P-Code VM.

Supported types: int, bool, set. Set capacity: 200 unique integers.

Scope: nested block scope with shadowing and duplicate declaration checks.

Core features: assignment, arithmetic, relations, boolean logic, if/else, while, read/write.

Set features: literal, add, remove, in, isempty, union, inter, write set.

Bonus 1: P-Code listing and single-step execution.

Bonus 2: set equality for set variables.

Bonus 3: set comprehension syntax: { expr | x in S if condition }.

Tests: make test-all passes positive tests test1-test5 and 8 negative tests.

See design.md and test-results.md for the complete Chinese report.